

Csaba Purszki

SENIOR C++ SYSTEMS ENGINEER

CONTACT

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Computer Vision Kft · VAT HU25100145

CORE SKILLS

Modern C++

STL, RAII, templates, cache-aware optimization

Low-Latency & Lock-Free

atomics, memory ordering, real-time hot paths

Concurrency Verification

reproducible race / memory-order tests

Performance Engineering

profiling, custom allocators, O(1) allocation

Architecture & Systems

ABI compatibility, IPC, cross-platform builds

EDUCATION

- Chaincode Labs BOSS — participant, 2026
- MSc Electrical Engineering, BME, 2000–2005

LANGUAGES

- Hungarian (native)
- English (fluent)
- German (basic)

OTHER

- Traditional musician
- Weekend sailor

PROFILE

Senior C++ systems engineer with over 20 years of experience in real-time, low-latency, and safety-critical software. My main technical areas are modern C++, concurrency, lock-free design, memory management, shared-memory IPC, and cross-platform architecture. I have designed reusable frameworks and system components for automotive ADAS products used in mass production. I am now applying this systems-level C++ expertise to Bitcoin protocol research and to new challenges in financial technology and distributed systems.

SELECTED ACHIEVEMENTS

- Custom Memory Allocator** — Custom allocator that avoids fragmentation and provides O(1) allocation, delivering deterministic performance in a mission-critical real-time system.
- Deterministic Shared-Memory IPC** — Low-latency, single-writer/multi-reader shared-memory queue for real-time ADAS surround view; design parallels concepts later adopted by iceoryx2.
- Patented Parking-Slot Detection** — Software architecture of a camera-based parking-slot-marking detection algorithm used in mass-production ADAS; contributed to a patent.
- C++ protobuf layer for nanopb** — A C++ layer over nanopb to make protocol-buffer integration easier in embedded projects.
- Light-Client Block Filter Research** — BIP 158 vs Binary-Fuse across 50,000 mainnet blocks and 10 wallet scenarios; Fuse16 kept zero false negatives while cutting client-side query CPU by 9–81× (desktop) and 6–46× (Raspberry Pi 5), at bandwidth similar to GCS.

EXPERIENCE

Valeo Vision Systems — Independent Contractor

05/2014 – Present

Computer Vision Kft · Budapest, Hungary / Ireland — Automotive CV

Algorithm Framework Expert

- Developed a reusable algorithm framework used by multiple teams for memory pools, debugging and visualization, logging, and shared-memory I/O.
- Provided technical leadership, mentored junior engineers, and introduced good practices for concurrency and resource management.
- Improved team efficiency by introducing test-driven development with GoogleTest.

Algorithm Embedded Lead — Parking Slot Marking Detection

- Designed the software architecture of the patented Parking Slot Marking Detection algorithm.
- Helped develop a package-manager-based build system using semantic versioning and CI/CD for stable multi-module integration.
- Made backward compatibility easier, reduced integration work, and simplified releases.

Valeo Vision Systems — Software Engineer

05/2011 – 05/2014

Tuam, Co. Galway, Ireland — Automotive CV

DSP Lead — Calibration and Scene Viewing Group

- Designed the software architecture of computer vision algorithms.
- Helped define development processes and code-reuse practices.

Algorithm Integrator — Software Group

- Integrated and maintained an automatic road-based camera calibration algorithm across customer projects.
- Refactored and ported the algorithm from C to C++ as a reusable library optimized for TI VisionMid and OMAP platforms.

Intellio Zrt — Algorithm Developer

08/2005 – 04/2011

Budapest, Hungary — Video Surveillance Systems

- Designed an algorithm development framework for object-tracker-based applications.
- Developed computer vision algorithms: object tracker, background subtraction, camera calibration, color recognition, shadow filtering.
- Developed a traffic detector using statistical traffic-flow analysis to identify stopped and wrong-direction vehicles.

PATENTS

Contributor to a German patent related to parking-slot detection (DE102019102561A1, "Verfahren zur Erkennung einer Fahrbahnmarkierung").